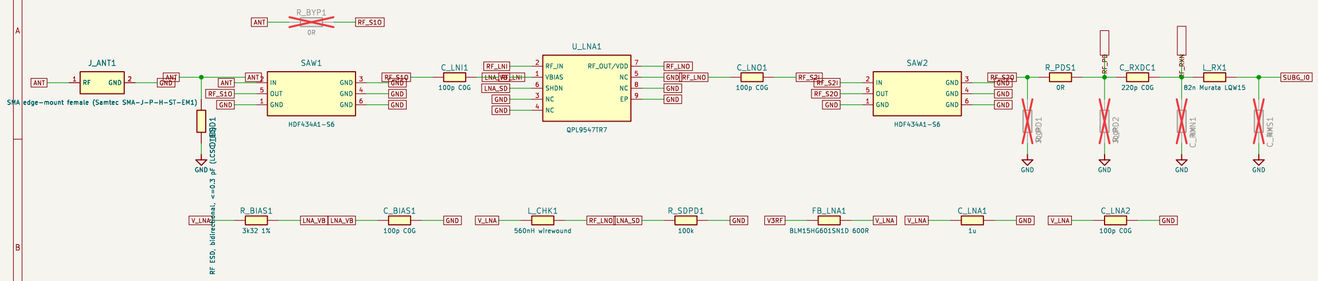
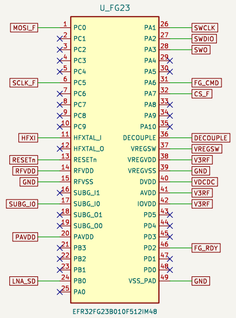
RF CHAIN 430–440 MHz: SMA – RF ESD – SAW1 – QPL9547 (~25 dB) – SAW2 – pi-pad option – RX match – EFR32FG23 SUBG_IO. Spec: SatHunter-FG23-OTG-nyak-spec-2026-11-07

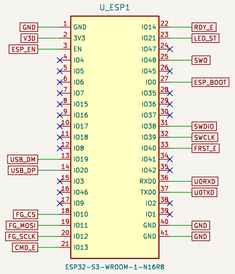
Gain to the FG23 ~21 dB, system NF ~2.6 dB (0.6 dB with SAW1 bypassed). Shield can over SAW1..FG23+TCXO (layout step).



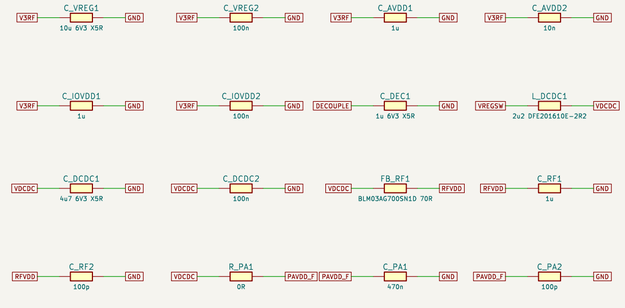
EFR32FG23 – RX only (link pins = ionos firmware: PC00 MOSI, PC05 SCLK, PA07 CS, PA06 CMD rx, PD02 RDY, PA03 SWO)



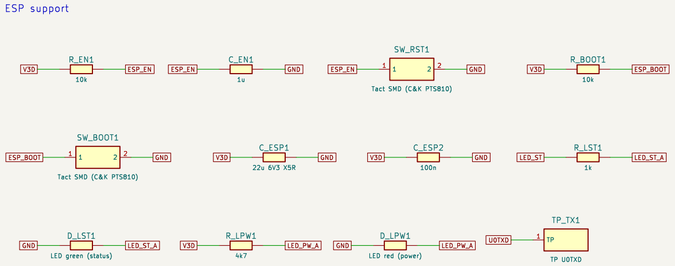
ESP32-S3-WROOM-1-N16R8 (SPI2 IO_MUX slave, TinyUSB CDC, SWD bit-bang to the FG23)



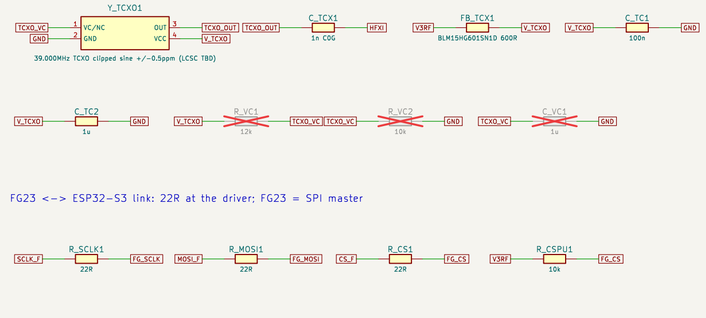
FG23 supplies (fg23-cw-trainer A5 / BRD4265B values; PAVDD from DC-DC as BRD4265B R211)



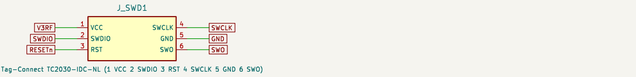
ESP support



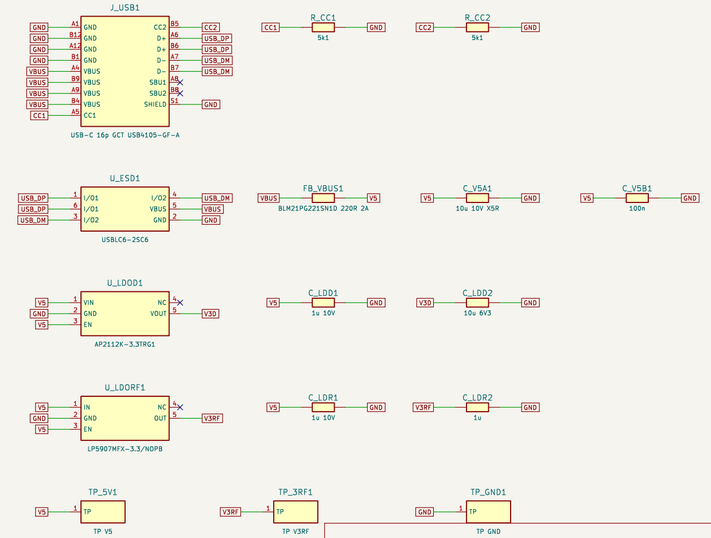
39 MHz TCXO -> HFXTAL_I (1 nF), HFXTAL_O open; Vc parts only for a VCTCXO



FG23 <-> ESP32-S3 link: 22R at the driver; FG23 = SPI master



USB-C (OTG device) + ESD + supplies: V5 -> AP2112K (ESP) / LP5907 (RF)



generated by generator/gen_sathunter.py – do not hand-edit; label-style, RF chain drawn explicitly

ANT-SAW-QPL9547-SAW-EFR32FG23 (RX) – ESP32-S3-WROOM-1 – USB-C OTG to the SatHunterSD

HA7DCD / Z2 Labs

Sheet: /

File: sathunter_fg23.kicad_sch

Title: SatHunter FG23 OTG – 70 cm satellite receiver

Size: A2	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1